Hubert Cadieux

Social Data Scientist

Education

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Master's in Political Science (Thesis track)

2024 - present
Université Laval, Quebec City

Courses and Learning:

- Master's thesis focusing on optimizing growth potential for political parties in multiparty systems
- Developing expertise in the following areas of political science:
  - Application of machine learning in political science
  - Application of Bayesian approach in political science
  - Quantitative analysis
  - Representation of political interests
  - Political communication
  - Political philosophy
- Deepening the research process in quantitative political science

Integrated Bachelor's in Economics and Politics

2020 - 2023
Université Laval, Quebec City

Courses and Learning:

- Honors mention from the Faculty of Social Sciences

- Completed the course *POL-2425 Persuasion and Public Opinion* which deepened my understanding of public opinion studies.
- Introduction and in-depth study of the main theories of political science and economics.
- Acquired fundamentals of research in political and economic sciences.
- Designed and developed a research project analyzing Université Laval students' perceptions regarding the economic aspects of Quebec City's tramway project, involving survey design, data collection and analysis, and presentation of results.

Diploma from the École interdisciplinaire outils & méthodes (EIOM)

2022 - 2023
Graduate School of International Studies (ESEI), Quebec City

Courses and Learning:

- Followed the path *Distinguishing Cause from Effect: Causality and Experimentation*
- Followed the path *Data Science: Modeling and Prediction*
- Mastery of advanced methods in analysis and modeling, including multiple regression, machine learning, data visualization, and dimensionality reduction.
- Acquisition of fundamental skills in logic and causal structuring.
- Development of skills in causal inference from observational data, applying techniques and methods such as synthetic groups, difference-in-differences, regression discontinuity, and matching.

Innovation and Potential Validation Program (VIP)

2024
Entrepreneuriat ULaval, Université Laval, Quebec City

Courses and Learning:

- Acquisition of basic accounting and financial forecasting concepts
- Development of an investor pitch

Business Opportunity Validation Program (VOA)

2024
Entrepreneuriat ULaval, Université Laval, Quebec City

Courses and Learning:

- Deepened my reflection on my business idea.
- Value proposition, competitive analysis.

AI Bootcamp

2023
Institut intelligence et données (IID), Université Laval, Quebec City_

Courses and Learning:

- Learning the fundamental principles of machine learning and deep learning, with practical application using scikit-learn and PyTorch.
- Exploration of concepts and challenges related to ethical and responsible AI.
- Introduction to optimization methods for machine learning models, including high-performance computing and Bayesian optimization.

College Diploma (DEC) in Social Sciences - administration profile

2017 - 2020
Cégep de Saint-Hyacinthe

Courses and Learning:

- Introduction to the main fields of social sciences.
- Acquisition of basic knowledge in linear algebra and differential calculus.

Work Experience

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Launching Opubliq

2024 - present

- Opubliq is a public opinion data anlysis startup I am currently launching.
- Design and prototyping of a **Software as a Service (SaaS)** adapted to the needs of municipal political parties.

- Active networking to establish strategic partnerships and expand the project's reach.
- Conducted market research to identify opportunities and refine product positioning.
- Funding applications for a startup
- Validation of hypotheses through user testing and stakeholder interviews.

- Development of a marketing and customer acquisition strategy.

Teaching Assistant

July 2025
Interactive Visualization, Dashboards, and Apps with R and Shiny, ICPSR Summer Program, Michigan State University, Ann Arbor

- Course setup in collaboration with Yannick Dufresne, course instructor.

- Supporting students during lab sessions.

Research Assistant

2021 - present
Leadership Chair in the Teaching of Digital Social Sciences (CLESSN), Université Laval, Quebec City

- Mastery of the R programming language, with direct application in specific theoretical contexts.

- Essential contribution to CLESSN's flagship projects: Datagotchi, Polimètre, and the Quorum Project.

- Refinement and application of advanced data analysis techniques to support research and publication projects.

- Co-writing and production of academic publications, detailed in the Publications and Conferences section.

- Active role in the design and implementation of software infrastructure (on AWS) and a data platform specifically adapted to the needs and challenges of digital social sciences.

- Preparation and submission of grant applications and ethics approval requests.

- Collaboration with external partners.

- For more details on specific projects and achievements, see the Achievements section.

Data Scientist

2022 - 2024
Mitacs, Musée de la civilisation de Québec (MCQ), Quebec City

- Enhancing data from surveys conducted by the MCQ.

- Application of advanced analysis techniques in a professional practical case.

- Writing and production of two strategic reports prescribing targeted marketing orientations for the MCQ.

- Active collaboration and communication with the MCQ marketing team.

Data Scientist

2023 - 2024
Interdisciplinary Center for Research on Africa and the Middle East (CIRAM), Université Laval, Quebec City

- Enhancement of a database on political dynasties.

Teaching Assistant

2022 - 2024
POL-7004 - Quantitative Analysis, Université Laval, Quebec City

- Grading exams and exercises.

- Technical assistance to students.
- Theoretical assistance to students.

Student Representative

2023 - present
Centre de recherche en données massives, Université Laval, Quebec City

- Represent student interests within the center.
- Participation in management committee meetings.
- Help organize social events.

Researcher for the Polimètre

2021
Center for Public Policy Analysis, Université Laval, Quebec City

- Monitor the fulfillment of the Trudeau government's promises (2019-2021).

Amateur Scout

2019 - 2021
Montréal-Est Rangers

- Attend Midget AAA games.
- Submit reports on interesting players.
- Preparation for the LHJAAAQ draft.

Teaching Assistant

2023
EIOM, ESEI, Quebec City

- Technical assistance to students.

Publications and Conferences

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Journal Articles

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Cadieux, Hubert, Catherine Ouellet, Sarah-Jane Vincent, Jérémy Gilbert, Camille Pelletier, and Yannick Dufresne. 2024. "Electoral Dynamics in Quebec: Analyzing the Rise of the Parti Québécois Through the Lens of Potential for Growth." *French Politics*, December. https://doi.org/10.1057/s41253-024-00274-z.

Déry, Axel, Alexis Bibeau, Hubert Cadieux, and Yannick Dufresne. 2022. "Sur Le Potentiel de Croissance Du Parti Conservateur Du Québec." *Canadian Journal of Political Science/Revue Canadienne de Science Politique* 55 (4): 972–77. https://doi.org/10.1017/S0008423922000762.

M. Foisy, Laurence-Olivier, Jérémie Drouin, Camille Pelletier, Jozef Rivest, Hubert Cadieux, and Yannick Dufresne. 2024. "Ain't No Party Like a GPT Party: Assessing OpenAI's GPT Political Alignment Classification Capabilities." *Journal of Information Technology & Politics*, December, 1–13. https://doi.org/10.1080/19331681.2024.2444587.

Thompson-Collart, Brian, Hubert Cadieux, Catherine Ouellet, and Yannick Dufresne. 2024. "Lessons Learned: Citizen Forecasting, Candidate Resignations, and the 2024 US Presidential Election." *PS: Political Science & Politics*, 1–16.

Tremblay-Antoine, Camille, Hubert Cadieux, Yannick Dufresne, and Adrien Cloutier. 2024. "Pledge Salience: A Scoping Review of a Non-consensual Concept."

Conference Presentations

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Cadieux, Hubert. 2023. "On Volatility and Parties' Potential for Growth in Multiparty Systems: The Case of the 2022 Quebec Election." St. Pete Beach, FL, USA.

———. 2024a. "Forecasting Election Swings in Multiparty Systems Using Underlying Potential for Growth of Parties." Nouvelle-Orléans.

———. 2024b. "Optimizing Electoral Strategies in Quebec's Multiparty System: A Machine Learning Approach to Assessing Party Potential for Growth." Québec (Québec).

Gajevic Sayegh, Alexandre. 2022. "Can Fairness and Green Jobs Accelerate the Energy Transition in Canada?" Montréal, Qc, Canada.

Media Articles

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Blais, Stéphane. 2022. "L'importance du climat varie selon l'allégeance politique." *Le Devoir*. https://www.ledevoir.com/politique/quebec/760221/l-importance-du-climat-varie-selon-l-allegeance-politique.

Descurninges, Clara. 2022a. "La polarisation en données: un portrait plus nuancé qu'on pourrait le croire." *Journal Métro*.

———. 2022b. "Legault évite de parler d'économie en ligne, mais domine quand même sur cet enjeu." *Le Soleil*. https://www.lesoleil.com/2022/09/17/legault-evite-de-parler-deconomie-en-ligne-mais-domine-quand-meme-sur-cet-enjeu-96785889a0cdd48da6cb3fda3fbe7c5a/.

Scholarships Received

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Master's Research Scholarship

2024 - 2026
Fonds de recherche du Québec (FRQ), Society and Culture

Next Generation Support Scholarship - Bachelor's

2021 - 2022
International Observatory on the Societal Impacts of AI and Digital Technology (OBVIA)

Achievements

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- Fusion of multiple political surveys.
- Creation of the R package *sondr* which offers useful functions for survey analysis.
- Creation of R packages useful for digital social science data analysis.
- Mentoring novice students in data analysis projects.

- Member of the Center for Public Policy Analysis (CAPP) and the Political Communication Research Group (GRCP).

- Training CLESSN students in the extract-transform-load (ETL) methodology in an academic context.

- Data analyst in the creation of the Datagotchi voting prediction tool, which attempts to predict individual voting based strictly on lifestyle habits.

- Data analyst for CLESSN's coverage of the 2022 Quebec election campaign in partnership with The Canadian Press.

- Contributions to Project Quorum – energy transition component: literature review on citizen attitudes and perceptions, design of an 80-question pilot survey, and market study for web development.

- Development of a Web application facilitating the organizational management of the Polimètre.

- Programming of a conjoint analysis on Qualtrics software to identify Quebecers' biases regarding different electoral bases.

- Layout work for the manuscript of the collective book *Assessment of the CAQ Government: Between Nationalism and Pandemic* by Birch et al., published in 2022.

- Analysis of Université Laval Foundation data.



Contact Information

hubertcadieux@opubliq.com hubert.cadieux.1@ulaval.ca hubertcadieux@gmail.com (438) 825-3958 6-70 Saint-Louis Street, Quebec City, QC, Canada

Languages

Programming Languages

Skills

- Data analysis
- Bayesian analysis

- Predictive modeling
- Local parallel computing
- Advanced data visualization
- Survey programming
- R package creation
- Mentoring, teaching
- Project management
- Coordination of ethics approval request